

Response to editor's and reviewers' comments

We thank the editor and two reviewers for their valuable comments that have helped to improve our manuscript. Below we give details of our response to these comments.

Comments from Associate Editor: Dr Manuela Gonzalez Suarez

1. Two reviewers have now assessed this manuscript. Both agree this can be a valuable contribution and also offer multiple suggestions for improvement that the authors should carefully consider for a revised version.

Response: We thank the editor and the reviewers for their positive assessment of the study.

2. In addition, both reviewers considered this manuscript did not lead to clear management recommendations (in one of the standard questions they are asked to respond when submitting a review). There are some general recommendations already in the conclusions but it would be great if the authors could add some more explicit management or policy-related recommendations derived from this study.

Response: We appreciate this suggestion by the reviewers and editor regarding management recommendations. We agree that it is critical to communicate these in order to shape how scientific evidence is used in conservation practice as well as how this process is studied. As a result, we have largely rewritten the conclusions section (see lines 591- 637) to provide more concrete actions that both researchers and decision-makers can take to improve the use of scientific evidence in conservation.

Reviewer 1

1. The authors have conducted a systematic map to identify what factors have been studied in terms of how they affect use of evidence by decision makers (i.e., policymakers and practitioners). I found this to be a clearly reported synthesis, the methods were transparent, and the findings are likely to guide the field of research in ways that help studies be more effective in understanding how to close the science-policy gap. I especially appreciated the focus and discussion on the interactions among factors, rather than considering them in isolation. I found the synthesis to be well discussed, the findings not overextended, and it should provide a clear roadmap for how people study the uptake of evidence in conservation policy and practice.

Response: We thank the reviewer for their encouraging words and are glad that they think this work could be useful.

2. I appreciated that the authors acknowledged the potential bias introduced by searching only in English and only including studies published in English. I think it would be worth a short discussion about how this may impact the number of studies found that do consider language barriers. For example, on lines 73-74 the authors cite language barriers as something that has been investigated, and I wonder if it has been investigated more in studies published in languages other than English.

Response: We thank the reviewer for noticing this potential bias that we overlooked – it's a good point. We have now included an acknowledgment of this in our section on biases and methodological limitations (see lines 580-583).

3. The methods were well reported and justified, including skipping critical appraisal (a sometimes nearly impossible task with conservation literature where the lack of reporting makes it hard to score) and given the scope of the search. The one thing that would be nice to have slightly more detail on is the Rayyan AI model and how reproducible its decisions are, for researchers considering similarly large syntheses. The use of a sensitivity check to estimate false positive / false negative rates was a great addition, but it would also be nice to have a bit more detail on the AI screener decision making process.

Response: This is a good point that we didn't consider before. Rayyan's AI classification model uses a machine learning classifier based on Support Vector Machine (Bolaños et al., 2024). This represents each abstract as a data point in a multidimensional space in which information from the abstracts is used in an attempt to define boundaries between different inclusion classifications, which is subsequently used to predict the inclusion of unclassified abstracts (Bolaños et al., 2024; Suthaharan, 2016). While the classifier requires a minimum of only two studies to be included and five to be excluded to allow for initial training of the model, its accuracy increases with a larger set of seed studies (Bolaños et al., 2024): hence why we initially screened 1000 studies. In terms of its performance, we found that it was much better at identifying studies to exclude than studies to include, in line with what other researchers have found (Valizadeh et al., 2022). We have included this information in the new version of the manuscript along with an explanation of how false negatives might have impacted our study (see lines 180-192, and lines 194-199).

4. It would also be good to clarify the difference between 'accessibility' and 'difficult to find' in the characteristics of the evidence. It is not clear from the table how those two things are different.

Response: We agree with the reviewer that the previous version of the manuscript did not effectively distinguish between accessibility and literature being difficult to find. We have now modified Table 1 to clarify these differences, with accessibility being related to whether people can access studies once it has been found (e.g. as a result of paywalls) while difficulty in finding literature relates to the inability to find studies when the search is undertaken (e.g. when using search platforms).

5. Line 234-235: I genuinely have no idea what this sentence was meant to say.

Response: We agree with the reviewer, reading it back we almost had no idea what it was meant to say either! We have now rewritten the sentence so it clearly describes the aims of the analyses we did (see lines 248-249).

6. Line 513: missing the word 'in' or 'withing' in this sentence.

Response: Thanks for spotting this. We have now corrected this sentence (see lines 562-565).

Reviewer: 2

7. This is an interesting study on a complex and worthwhile topic and which provides important evidence for the ongoing debates and progress around evidence-based conservation and decision making, including in policy and practice. These analysis and data are valuable since there is generally a poor and non-standardized documentation on what are the factors influencing the inclusion (or not) of evidence and thus ways to improve this.

Response: We thank the reviewer for their encouraging words and for acknowledging the value of our study.

8. The authors use a systematic approach for producing and reporting of the systematic map methodology and describe the process well which is important given the depressingly large number of poor, biased and opaque reviews that continue to be published on various topics.

Response: We thank the reviewer for acknowledging our hard work to make our study as robust and transparent as possible.

9. However, I think that the text needs thorough and detailed editing to fully use and communicate the main findings and especially the practical implications of this study.

Response: We appreciate this comment and acknowledge the importance of communicating the practical implications of this study for researchers, practitioners, and policymakers. We have now rewritten the conclusions section to highlight more concrete actions that could be taken by scientists and decision-makers to improve the use of scientific evidence in conservation decision-making (see lines 591 – 637).

10. Abstract, line 6 “many management and policy decisions do not incorporate scientific information.” They may or may not but probably the issue is they do not do so explicitly? I would argue that many would say they do but if not done transparently it makes it very difficult to understand what is the basis for any statements or recommendations.

Response: We agree with the point about it being difficult to track the use of scientific evidence in conservation practice and that in certain settings, such as policy, evidence is often used without being acknowledged. However, there is also a large evidence base that suggests that scientific evidence is typically used much less commonly than personal experience or that of colleagues (Bayliss et al., 2012; Cook et al., 2010; Cvitanovic et al., 2014; Matzek et al., 2014; Pullin et al., 2004; Young & Van Aarde, 2011). We have modified the text in the abstract to reflect the concerns of the reviewer (see lines 2-3).

11. Lines 9-11- I would rephrase this as it seems unclear and general.
decisions and topic are so broad as to make it difficult to understand what is exactly the scope here

Response: We have now modified this text so that what we were trying to say is clearer (see lines 6-8).

12. Lines 12-14 same thing here

Response: We have now altered this text so that what we were trying to say is clearer (see lines 9-12).

13. Line 17- I think this statement is potentially misleading in the absence of an explanation to say only English-language studies were included in this systematic map.

Response: We acknowledge this may be an issue and have altered the text in the discussion (lines 415 – 420) to reflect this.

14. Lines 28-33 – Personally, I find these statement rather weak and very general. I think it would make a much stronger ms if they could be rewritten with more specificity and focus.

Response: We have now overhauled this section. It now presents much more concrete specific actions both researchers and decision-makers can take to aid the integration of scientific information (see lines 24-33).

15. Lines 63-67 Minor point but I think this could do with rephrasing as it suggests it's a solution to the whole issue of research action gap but it's mostly on prioritising what research is needed.

Response: We understand the point that the reviewer is making here, but we also see coproduction as much broader in its impacts on evidence use than simply the production of relevant evidence. Successful co-production helps to build relationships between stakeholders, identifies relevant topics, and delivers evidence in a format that suits user needs. We agree that it is not a solution to all the problems, but it does aim to deal with many of the issues relating the characteristics of scientific evidence. We have now added some text to clarify that co-production is just one of many strategies that could be used to increase the use of scientific evidence in conservation practice (see lines 64-69).

16. Line 81 I think this needs some details as otherwise surely everyone has time constraints

Response: Yes, this is a fair point. We have now added some text to clarify what we mean here (line 81-84).

17. Line 180 a verification rate of 5% seems very low given the implications of excluding such a vast number of papers, not far off 50% of the entire retrieved dataset. Not much that can be done at this stage but I would suggest putting in a clear caveat in relation to this aspect.

Response: While we agree with the reviewer that it would have been preferable to screen a larger number of papers that were automatically excluded as a sensitivity test, we lacked the resources to be able to do this. Extrapolating from our sensitivity analysis, we expect that automated exclusion likely led to around 20 potentially relevant primary studies being excluded at the title and abstract screening stage and around 4 relevant studies being excluded from the final synthesis. Previous studies have shown that automated exclusion is typically highly accurate with an ability to identify 95-99% of irrelevant literature (O'Mara-Eves et al., 2015; Valizadeh et al., 2022), whilst, on average, reducing the time needed for screening by 40% (Ouzzani et al., 2016). We have modified part of our methods section to reflect this caveat (see lines 189-199).

A flipside to this is that most syntheses of the primary literature in ecology and conservation, such as meta-analyses and systematic maps, use small searches that have high specificity (i.e. return a high proportion of relevant studies) but that have low sensitivity (i.e. return a low number of studies), resulting in them being likely to miss a large proportion of relevant studies in the literature. Our approach, while not perfect, is much less likely to face such problems due to the broad nature of our search.

18. Line 234 any biases?

Response: Thanks to the reviewer for spotting this. We have now edited this section so that what we were trying to express is clearer (see lines 248-249).

19. Line 258-259 isn't there a point also that they are some of the most research intensive countries in terms of number of universities and conservation science institutions and therefore a bias on evidence use in conservation is to be expected?

Response: Yes, this is a good point. We have added a sentence to our discussion to reflect this caveat (see lines 417-420).

20. Figure 1 and lines 272-278 given the small numbers overall I strongly suggest you include numbers as well as percentages for all subcategories. As a side note, quite extraordinary and depressing to see that most studies did not specify either ecosystem nor threat (60%!). These proportions are so high as to require careful explanations in the discussion. Was there a mismatch in the type of studies you retrieved and the typical ecological study?

Response: Thank you for this suggestion, we have now added numbers of studies for all the subcategories described in the text (see lines 286-293). Regarding the large number of studies that did not report ecosystem or threat type, we think the reasons for this are more subtle than purely researchers failing to report this information. The typical study used interviews or questionnaires to investigate how practitioners in a set of organisations used evidence and what factors influenced this. However, often the focus of these organisations was not conservation of a specific type of ecosystem or a particular type of conservation problem – many organisations managed multiple ecosystem types and dealt with multiple threats. In these cases, it was common not to report the details of the ecosystems or threats that organisations were dealing with. We have reflected on this and added a new section describing our thoughts as well as calling for researchers to publish more detailed information on the contexts in which studies were taken (see lines 427-438).

21. I really like Figure 2. Very simple, clear and easy to interpret. well done to the authors!

Response: Thank you for the positive feedback, we appreciate it!

22. Figure 4 is interesting as it shows a promising rapid uptake of such studies post 2000 and 2015 but your results indicate that the field is far from maturity, with relatively few studies overall and most of them not including even basic type information such as relevant ecosystem or threat.

Response: We have sought to address this issue with a new section that explains why we think many studies do not report information on ecosystem types or conservation threat as well as the need to report this type of information in more detail (see lines 427-438).

23. Discussion I would present the discussion points in the exact order of results to make it easier to follow.

While we understand the point the reviewer was making here, the discussion structure aimed to highlight what we think are the most important results – namely the frequency with which different factors have been mentioned in the literature. As such we have chosen not to change the order of the different paragraphs in the discussion.

24. Lines 384- see my earlier comment about research intensive institutions and not just universities (e.g. Birdlife is a global partnership but based in the UK)

Response: We have now added a caveat to this section relating to the number of research-intensive organisations found in these countries and the potential for this to be a driver of the geographic patterns we see (see lines 417 - 420).

25. Line 388 I think this statement about the Global South is problematic in the absence of any supporting data. I would rephrase or delete.

Response: We have slightly rephrased this section to make the uncertainty around the type of factors that might impact the use of scientific evidence in the Global South. However, we have decided to keep the rest of the section, rather than deleting since we think that expanding research to understudied regions is an important next step for research on evidence use (see lines 422-425).

26. Line 411-415 sorry, I am not clear what is the point here and it seems rather contradictory. Data generated by models and citizen science being of doubtful rigour and thus a factor limiting uptake of evidence?

Response: Sorry for not expressing clearly what we wanted to say here. What we were trying to say is that many studies highlight the importance of highly rigorous scientific studies, but that when data is generated using models or collected by citizen scientists, this data is often perceived as less trustworthy. We have now modified this section and split it into two sentences to make things clearer (see lines 457-462).

27. Line 449-451 I understand the second part but not sure what the researcher's knowledge of policy or practice contexts means or how it was quantified.

Response: We're sorry that this was not clear. We have rephrased this section so that it is easier to understand (see lines 496-500).

28. Line 502- yet we have very little information on the conservation costs, something which could presumably also influence the take-up of evidence into policy and practice. This might be worth mentioning as an important ongoing issue.

Response: In principle we agree with the reviewer here, there is definitely a lack of information about the costs of conservation actions. However, we were trying to make a more general point here that conservation is under-resourced. We have added an additional sentence relating to the potential burden imposed by the need for conservation actions to be based on scientific evidence which touches on the comment of the reviewer (see lines 553-556).

29. Line 513-514 Research always encourages more research but isn't this problematic given the current state of play you highlight (i.e. few studies and with most missing critical bits of information)? Would it not be more realistic to wait until the field is more mature or conduct more primary type research on these factors rather than syntheses to understand these factors?

Response: While we recognise that the number of studies on scientific evidence for a particular context is relatively small, we do not agree that 167 primary studies constitute a small number of studies. A key factor to consider is how consistent the factors that impact scientific evidence use are and whether they vary markedly between different contexts. If they are relatively similar across different contexts then there may be little need to repeat similar studies in different contexts. However, if they differ from one context to another, then synthesis could help to identify priority

knowledge gaps to fill. We suspect that the importance of some factors impacting evidence use will be consistent (e.g. relationships between researchers and actors), while others will vary depending on context (e.g. timeliness of evidence). We have now modified this section of text to more clearly reflect our feelings on potential future evidence synthesis on the topic (see lines 562-565). We have also added recommendations about potential topics and reporting in primary studies in other sections of the manuscript (lines 434-439, 592-606)

30. Line 514-519 I agree and I think this is an important consideration. It would be good to see a clear solution proposed for dealing with this in future research. A better, more robust expert solicitation?

Response: We have now modified this text slightly to clarify the issue and added a number of proposed solutions to the problem (see lines 566-573).

31. Line 538- this guidance element therefore seems very important but I think it is not explicitly phrased currently as such in terms of what exactly should researchers do in relation to the findings of this systematic map. If you refer to it as such I think it would be useful to rephrase some of the conclusions to make them more directly in the form of some specific next steps to incorporate.

Response: We agree with the reviewer that this section needed more explicit wording so that it is clearer what the next steps should be for researchers, practitioners, and policymakers. As such we have largely rewritten the conclusions section to provide more concrete suggestions (see lines 591 – 632).

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